



NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA-769008
MID-SEMESTER EXAMINATION, 2023

SESSION: 2023 – 2024 (Spring)
B. Tech. 8th / M. Tech. 2nd Semester

Subject code: CS6404
No. of pages: 2

Subject Name: Software Testing
Full Marks: 60

Dept. Code: CS
Duration: 2hrs

Figures at the right hand margin indicate marks.
All parts of a question should be answered at one place.
Answer all questions.

Q. No.	Particulars	Marks
1.		
(a)	A software was tested using error seeding strategy in which 40 errors were seeded in the code. When the code was tested using the complete test suite, 32 of the seeded errors were detected. The same test suite also detected 100 non-seeded errors. What is the estimated number of latent errors in the software after this testing is over?	2*10=20
(b)	Suppose, there are 3 input variables in a module. How many test cases can be designed using worst-case testing method?	
(c)	At least how many test cases are required to achieve MC/DC coverage of the following code segment if((a>10) OR (b<100)) AND ((c<50) OR (d>20)) then x = x+1;	
(d)	Give a positive and a negative test scenario for the “mobile number” field of the Student Registration Page in NIT Rourkela website.	
(e)	Suppose a program contains 5 decision points, each of which has 2 branches. How many test cases are necessary for branch testing?	
(f)	What is the relationship between cyclomatic complexity and program comprehensibility? Can you justify why such an apparent relationship exists?	
(g)	If there are 4 numbers of variables that have 3 values, then how many test cases are required to perform orthogonal array testing? Justify your answer.	
(h)	What do you mean by “ku” anomaly? What is its effect in data flow testing?	
(i)	If in a decision table, there are 3 rules and 4 binary conditions, then how many minimum and maximum test cases can be designed?	
(j)	What is JCUTE? Why it is used?	
2.		
(a)	Consider the following payroll system of an organization: (i) If the salary of an employee is less than or equal to Rs. 70,000 and expenses do not exceed Rs. 30,000, then 10% of the salary is deducted as income tax. (ii) If the salary is greater than Rs. 70,000 and less than or equal to Rs. 2,00,000 and expenses do not exceed Rs. 40,000, then 20% of the salary is deducted as income tax. (iii) For salary greater than Rs. 2,00,000, 5% additional surcharge is also charged, besides the income tax. (iv) If expenses are greater than Rs. 40,000, then 7% additional surcharge is also charged, besides the income tax. Design test cases for the above payroll system using cause-effect graph based technique.	12+8=20

(b) 3. (a) (b)	Consider a scenario in which we need to derive the test cases for a Banking Information System, that has four different functionalities as mentioned below: (i) Deposit money (ii) Withdraw money (iii) Transfer money (iv) Check balance These four functionalities can individually be <i>available</i> or <i>unavailable</i> or <i>partially available</i> to the users. Design test cases for the above Banking Information System using orthogonal array testing. Write a program in C for searching an integer value from a large stored sequence of integer values stored in an array of size 100, using linear search method. (i) Draw the CFG for the above program. (ii) Calculate the cyclomatic complexity of the program. (iii) List all the independent paths. (iv) Design all the test cases from the independent paths. Find out 4 possible mutants in the program developed according to Q 3(a). Check how many of them are killed by taking a set of arbitrary test data. Add new test data if required, to improve the mutation score.	12+8=20
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